

# Joonsuk Bae

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*ALICE PWG-JE Run 3 charged-particle jets — end-to-end pipeline for ALICE's first Run 3 inclusive charged-particle jet cross-section measurement at  $\sqrt{s} = 13.6$  TeV (preliminary; paper in preparation; EPS-HEP 2025 PWG-JE merge talk on behalf of ALICE). Pb/scintillating-fiber calorimetry R&D for the ePIC Barrel Imaging Calorimeter.*

## Education

**Ph.D. Candidate, Physics** — Sungkyunkwan University 2022.03 – expected 2027.02

Advisor: Beomkyu Kim. Thesis: *R-dependent charged-particle jet production in pp collisions at  $\sqrt{s} = 13.6$  TeV with ALICE*. Defense expected 2026.12; degree conferral 2027.02.

**B.S., Physics** — Sungkyunkwan University 2021.02

## Appointments and Collaboration Membership

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|----------------------------------------------------|-------------------|
| ALICE Collaboration, CERN, Geneva                  | 2022.10 – present |
| ePIC Collaboration, Brookhaven National Laboratory | 2024.02 – present |
| LAMPS Collaboration, RAON, Korea                   | 2022.03 – 2024.04 |

## Research Experience

**Main analyzer, ALICE Run 3 charged-particle jets (PWG-JE)** — Sungkyunkwan University / CERN 2023.04 – present

- **First ALICE Run 3 inclusive charged-particle jet cross section** at  $\sqrt{s} = 13.6$  TeV — responsible for the full chain: O<sup>2</sup>Physics task development, QA on continuous-readout data, Bayesian/SVD unfolding, and the systematic-uncertainty program. PWG-JE-approved preliminary results; paper in preparation. Varied  $R = 0.2$ – $0.6$  to separate collinear from large-angle radiation.
- **Tracking-efficiency systematics for Hard Probes 2026 (PWG-JE)** — provided the pp charged-particle tracking-efficiency systematics at  $\sqrt{s} = 13.6$  and  $5.36$  TeV ( $p_T$ -binned, 2022–2024 periods); the ITS-TPC matching code was reused by a PWG-JE colleague as a starting point for PbPb tracking studies.
- **Detector-level corrections for the new readout** — quantified the bunch-crossing and track-matching inefficiencies arising from continuous TPC readout and time-frame-based asynchronous reconstruction, and derived the corresponding signal-loss corrections; studied data–MC track  $p_T$ -resolution mismatch by covariance propagation, MC residuals, and V0 mass resolution, and derived the  $p_T$ -dependent smearing profile used in Run 3 charged-jet systematics.
- **Cross-section normalization** — implemented the Run 3 charged-jet normalization from the ALICE TVX visible cross section under continuous readout, now used at particle level by the SKKU Run 3 charged-jet analyses (pp 13.6 TeV and O+O); provided the normalization factor and QA for the O+O charged-jet measurement carried out by two MSc co-analysts I mentor.
- **Event-generator production for model comparison** — managed multi-setup POWHEG-BOX and PYTHIA 8 productions (Monash, colour-reconnection modes, rope hadronization, string shoving) plus HERWIG 7 samples for the Run 3 jet-radius comparison, with provenance tracking across configurations.

**Detector physicist, ePIC Barrel Imaging Calorimeter (BIC)** — Korea-BIC group 2024.02 – present

- **Pb/scintillating-fiber prototype test beams** — lead analyst for the CERN PS T10 (2025.07) campaign: energy response, resolution, and linearity. On-site DAQ shifter with equalization and prompt analysis at KEK AF-AR (2025.03); module performance characterization for the first PS T10 campaign (2024.08), published as a co-author.
- **Module QA and assembly** — PMT-module QA before the test beams (cosmic-ray and source characterization, timing-coincidence checks, optical-coupling stability, per-module acceptance criteria); Pb/scintillating-fiber bundle alignment, polishing, and PMT/SiPM coupling.

- Energy- and  $\eta$ -dependent sampling-fraction studies in JANA2 / EIC-recon for the Korea-BIC simulation effort; Geant4 prototype modeling.

## Detector R&D, LAMPS Forward Tracker — RAON, Korea

2022.03 – 2024.04

- Early-stage design and Geant4 simulation of an MPPC–scintillator-plane forward tracking detector; sensor surveys, specifications, and tracking-concept feasibility studies.

## Publications

**i** INSPIRE: [inspirehep.net/authors/2117232](https://inspirehep.net/authors/2117232) · ORCID: 0009-0008-4806-8019. ALICE Collaboration co-author since 2022.10.

### In preparation

- ALICE Collaboration, *R-dependent charged-particle jet production in pp collisions at  $\sqrt{s} = 13.6$  TeV with ALICE (in preparation). First ALICE Run 3 inclusive charged-jet publication; thesis paper.* [main analyzer; analysis, corrections, unfolding, systematic uncertainties; primary author of the internal analysis note and paper draft]

### Submitted (detector)

- Y. Hong, J. Bok, G. An, **J. Bae et al.**, “Test-beam performance of the AstroPix silicon sensor for imaging calorimetry,” submitted to *Nucl. Instrum. Meth. A* (NIMA-D-26-00572). Corresponding: Prof. Sanghoon Lim. [co-author; ePIC BIC silicon-layer test-beam analysis]
- H. Lee, C. Lee, J. Ryu, G. An, **J. Bae et al.**, “Beam test of a Pb/SciFi prototype for the Barrel Imaging Calorimeter at the Electron-Ion Collider” — CERN PS T10 (2024.08) campaign; submitted to *Nucl. Instrum. Meth. A* (NIMA-D-26-00482); arXiv:2604.22647. [co-author; module-construction support and PMT/module performance characterization (KPS Spring 2025 poster)]

### ePIC/EIC and related calorimeter R&D

- S. Y. Jang, G. An, **J. Bae et al.**, “Performance of dual-readout calorimeters for various absorbers,” *Nucl. Instrum. Meth. A* (2026), DOI 10.1016/j.nima.2026.171598.
- J. K. Adkins et al. (ECCE Collaboration), “Design of the ECCE detector for the Electron-Ion Collider,” *Nucl. Instrum. Meth. A* **1073** (2025) 170240. [co-author; ECCE consortium member]
- G. Cho, S. Kim, **J. Bae et al.**, “Beam tests of the copper-based dual-readout calorimeter to measure electromagnetic performance for future  $e^+e^-$  colliders,” *J. Subatom. Part. Cosmol.* **3** (2025) 100021. [co-author; module construction and beam-test data taking]
- ECCE Consortium, “AI-assisted optimization of the ECCE tracking system at the Electron-Ion Collider,” *Nucl. Instrum. Meth. A* (2023), DOI 10.1016/j.nima.2022.167748. [co-author; ECCE consortium member]
- ECCE Consortium, “Open heavy-flavor studies for the ECCE detector at the Electron-Ion Collider,” arXiv:2207.10632 (2022). [co-author; preprint]

### Earlier detector R&D (LAMPS, RAON; peer-reviewed)

- Y. J. Kim, D. S. Ahn, J. K. Ahn et al. (LAMPS), “Large Acceptance Multi-Purpose Spectrometer (LAMPS) at the RAON,” *J. Korean Phys. Soc.* **87** (2025) 670–682. [co-author; Time-of-Flight detector maintenance and on-site checks]
- Y. Bae et al. (LAMPS), “Performance test of prototype LAMPS Start Counter using proton beams at 100 MeV,” *Nucl. Instrum. Meth. A* (2025), DOI 10.1016/j.nima.2025.170827. [co-author; Start-Counter prototype testing]
- Y. Cheon et al. (LAMPS), “Characteristics of the prototype LAMPS TPC for low-energy nuclear collisions at RAON,” *Nucl. Instrum. Meth. A* **1066** (2024) 169610. [co-author; TPC prototype characterization]

- H. Kim *et al.* (LAMPS), “Performance of the prototype beam drift chamber for LAMPS at RAON with proton and  $^{12}\text{C}$  beams,” *JINST* **19** (2024) P12008; arXiv:2412.08662. [co-author; forward-tracking R&D and prototype performance studies]

## Software and Computing

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### ALICE O<sup>2</sup>Physics — AliceO2Group/O2Physics

- Contributor since 2023: jet-finding and response-matrix tasks, systematic-uncertainty pipelines, particle-level cross-section normalization, and shared track-smearing tools.

## Invited and Contributed Talks

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**Summary since 2022:** 3 invited workshops [external] · 5 invited collaboration-meeting talks [internal] · 8 international conference contributions (one on behalf of the ALICE Collaboration at EPS-HEP 2025; a parallel talk and a poster at Hard Probes 2026) · 9 domestic (KPS) · regular PWG-JE working-group speaker since 2023.

### Invited Talks (external workshops, seminars)

- **France–Korea Particle Physics Network (FKPPN) Workshop**, Inha University — *Cross-section normalization in Run 3; charged-jet studies in ALICE Run 3* — 2025.12
- **1st Heavy-Ion Meeting (HIM 2025)**, APCTP — *From precision to structure: jet measurements in ALICE Run 3 and ongoing substructure studies* — 2025.05
- **FKPPN Workshop**, Inha University — *Charged-jet studies in ALICE Run 3* — 2023.11

### Collaboration-Meeting and National-Workshop Talks (internal)

- **Korea-BIC 2025 Summer Workshop** — *Data analysis of the recent beam test at CERN* — 2025.09
- **KoALICE National Workshop 2024** — *Jet analysis status: first look at the charged-particle jet spectrum in Run 3 pp collisions* — 2025.01
- **ALICE Physics Week 2024** — *Charged-particle jet production in Run 3* — 2024.12
- **KoALICE National Workshop 2023** — *Charged-jet O<sup>2</sup> MC and data QA* — 2024.01
- **KoALICE National Workshop 2022** — *Dijet studies with ALICE at the LHC* — 2023.01

### Contributed Talks and Posters

- **Hard Probes 2026**, Nashville — *Jet-like dihadron correlations in pO, OO, and Ne–Ne collisions with ALICE* [international, talk] — 2026.06
- **Hard Probes 2026**, Nashville — *Characterizing inclusive charged-particle jet production and fragmentation in pp collisions* [international, poster] — 2026.06
- **EPS-HEP 2025**, Marseille — *Inclusive and semi-inclusive jet cross-section measurements in pp collisions at  $\sqrt{s} = 13.6$  TeV with ALICE* [on behalf of the ALICE Collaboration; PWG-JE merge talk] — 2025.07
- **INPC 2025**, Daejeon — *Charged-particle jet and jet quenching measurement in pp collisions at  $\sqrt{s} = 13.6$  TeV during LHC Run 3 with ALICE* [international, talk] — 2025.05
- **Quark Matter 2025**, Frankfurt — *Inclusive charged-particle jet spectrum in pp collisions in Run 3 at  $\sqrt{s} = 13.6$  TeV with ALICE* [international, poster] — 2025.04
- **Hard Probes 2024**, Nagasaki — *First measurements of charged-particle jet production in pp collisions at  $\sqrt{s} = 13.6$  TeV with ALICE* [international, poster] — 2024.09
- **Quark Matter 2023**, Houston — *Charged-particle multiplicity distribution in pp collisions at  $\sqrt{s} = 13.6$  TeV with ALICE* [international, poster] — 2023.09
- **ATHIC 2023** (9th Asian Triangle Heavy-Ion Conf.), Incheon — *Dijet studies at the LHC* [international, poster] — 2023.04
- **KPS Spring 2026**, Daejeon [domestic, talk] — *R-dependent charged-particle jet production in pp collisions at  $\sqrt{s} = 13.6$  TeV with ALICE* — 2026.04

- **KPS Fall 2025**, Daejeon [domestic, talk] — *Inclusive and semi-inclusive jet cross-section measurements in pp collisions at  $\sqrt{s} = 13.6$  TeV with ALICE* — 2025.10
- **KPS Spring 2025** [domestic, talk] — *Inclusive charged-particle jet spectrum in pp collisions in Run 3 at  $\sqrt{s} = 13.6$  TeV with ALICE* — 2025.04
- **KPS Spring 2025** [domestic, poster] — *Prototype calorimeter QC for the Barrel Imaging Calorimeter at the EIC* — 2025.04
- **KPS Fall 2024** [domestic, talk] — *First jet measurement in ALICE Run 3: charged-particle jet production* — 2024.10
- **KPS Spring 2024** [domestic, talk] — *Charged-particle jet production in pp collisions at  $\sqrt{s} = 13.6$  TeV* — 2024.04
- **KPS Fall 2023** [domestic, talk] — *First look at charged-particle jet production in pp collisions at  $\sqrt{s} = 13.6$  TeV with ALICE Run 3 data* — 2023.10
- **KPS Spring 2023** [domestic, poster] — *Dijet studies at the LHC* — 2023.04
- **KPS Fall 2022** [domestic, poster] — *Dijet studies with ALICE at the LHC* — 2022.10

## Test-beam and Detector Activities

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### ALICE ITS2 alignment monitoring at CTF level (Run 3)

- Analyzed compressed-time-frame (CTF) data — below the standard AO2D analyzer interface — to provide period-by-period tracking-performance diagnostics.

### ePIC Barrel Imaging Calorimeter test beams

- **CERN PS T10** (2025.07) — lead analyst; energy response and resolution.
- **KEK AF-AR** (2025.03) — DAQ, equalization, prompt analysis.
- **CERN PS T10** (2024.08) — first campaign; module performance characterization.
- **Korea-BIC module production** (2024 – 2025) — Pb/scintillating-fiber bundle assembly, polishing, and PMT/SiPM integration; PMT-level QA on the prototype modules used in the 2025 beam-test campaigns.

## Service and Professional Activities

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- **PWG-JE tracking-efficiency contribution for Hard Probes 2026** — pp 13.6 TeV studies; ITS-TPC matching code reused by a PWG-JE colleague as a starting point for PbPb tracking studies.
- **ALICE Run 3 O+O charged-jet measurement** (PWG-JE) — co-author on the Analysis Note; service contribution: cross-section normalization factor derivation and QA; co-mentor of the two MSc analyzers.
- **ALICE ITS2 alignment monitoring** — service work at CTF level for tracking-performance diagnostics.
- **ePIC Korea-BIC analysis coordination** — analysis liaison between the SKKU group and the Korea-BIC group for beam-test data products.
- **KoALICE national activities** — regular participation since 2022, including invited collaboration-meeting talks.

## Awards and Fellowships

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- **BK21 FOUR Graduate Scholarship** — Sungkyunkwan University Physics BK21 sponsored unit (Korean Ministry of Education), 2022 – present.
- **Conference travel support** — KoALICE, NRF Korea–CERN program, and SKKU Graduate School; HP 2024 (Nagasaki), QM 2025 (Frankfurt), EPS-HEP 2025 (Marseille).

## Teaching and Mentoring

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**Student mentoring (non-supervisory)** — Sungkyunkwan University 2024 – present

- *Wooseok Ham* (M.Sc., co-analyzer on the ALICE Run 3 O+O charged-jet measurement) and *one further MSc co-mentee* (co-analyzer on the same O+O analysis) — mentored on Run 3 jet reconstruction methodology, correction procedures, and systematic-uncertainty estimation; co-author with W. Ham on the corresponding

PWG-JE Analysis Note.

- *Harim Seong* (undergraduate) – ePIC BIC detector-performance studies using JANA2-based reconstruction; energy response and sampling-fraction extraction.

**KoALICE O<sup>2</sup> Tutorial – student co-organizer and instructor** – Sungkyunkwan University, Suwon 2024.01.22–23

- Co-organized (with Hyungjun Lee) the two-day Run 3 analysis-framework tutorial for the Korean ALICE community – ~22 participants from SKKU, Inha University, and other Korean institutes; listed as the official Indico contact for the workshop. [indico.cern.ch/event/1361235](https://indico.cern.ch/event/1361235)
- Instructor on *Running O<sup>2</sup>Physics with Hyperloop* (solo) and co-instructor on *Practical PWG-JE tasks* (with H. Lee).

**Teaching Assistant**, SKKU Physics (2022 – 2024) – *Nuclear Reactions* (graduate), *Quantum Mechanics II*, *General Physics I and Laboratory I*.

## Technical Skills

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**Languages.** English (professional working proficiency), Korean (native).

**Programming.** C++, Python, ROOT, L<sup>A</sup>T<sub>E</sub>X, Bash, Git.

**HEP frameworks.** ALICE O<sup>2</sup>Physics, Geant4, JANA2 / EIC-recon, FastJet.

**Event generators.** PYTHIA 8 (Monash, colour-reconnection modes, rope hadronization, string shoving), POWHEG-BOX, HERWIG 7.

**Analysis methods.** jet reconstruction and unfolding (Bayesian, SVD), efficiency and systematic-uncertainty evaluation, data/MC validation, test-beam calibration and energy-response fitting.

**Detector hardware.** PMT and SiPM readout characterization, cosmic-ray and source calibration, optical-coupling QA, calorimeter module assembly.

## References

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Available on request.